SYNTHETIC TEST DATA ? NOT AN OFFICIAL AUTOSAR DOCUMENT
Document ID: AUTOSAR-HLD-SYNTH-2026-V1 | Revision: 1.0 | Status: Draft Engineering Specification
Title: High-Level Design (HLD) Document ? Automotive Powertrain & Body Domain Controller

Section 1: Executive Summary & Architectural Scope

This document specifies the High-Level Design (HLD) for the next-generation Electric Vehicle Powertrain Domain Controller (PDC) and Body Control Module (BCM). The system is developed in accordance with the AUTOSAR Classic Platform Release 4.4.0 specification.

1.1 Purpose and Architectural Objectives

The primary purpose of the Powertrain Domain Controller is to manage vehicle propulsion, torque coordination, high-voltage battery state management, thermal cooling loops, and high-speed gateway communication. The software architecture strictly enforces modular component separation through the AUTOSAR Virtual Functional Bus (VFB) and the Runtime Environment (RTE).

1.2 Layered Architecture Overview

The system follows standard AUTOSAR layered architecture:
- Application Software Layer (ASW): Houses Atomic Software Components (SWCs).
- Runtime Environment (RTE): Facilitates location-independent inter-SWC communication and Basic Software abstraction.
- Basic Software (BSW): Includes Microcontroller Abstraction Layer (MCAL), ECU Abstraction, System Services (EcuM, BswM), and Communication Stack (Com, CanIf, CanSM, PduR).
All interfaces conform to AUTOSAR Sender-Receiver and Client-Server communication paradigms.

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Section 2: Architectural Overview & AUTOSAR Layered Model
The architecture decomposes vehicle functions into distinct, decoupled software components (SWCs). Inter-component communications are mapped through standardized AUTOSAR Port Prototypes.

2.1 Execution Model
The Powertrain Domain Controller executes on a multi-core 32-bit automotive microcontroller (e.g., TriCore AURIX or RH850). The operating system is an OSEK/VDX compliant AUTOSAR OS supporting static priority-preemptive task scheduling:
- Task_10ms: High-priority periodic task executing powertrain throttle and brake torque loops.
- Task_20ms: Medium-priority task managing battery pack voltage and current monitoring.
- Task_100ms: Low-priority supervisory task coordinating thermal loops and gateway routing.
- Task_Background: Non-time-critical diagnostic and background memory verification.

2.2 Memory Mapping and Safety Isolation
Components assigned to safety-critical functions (ASIL-C / ASIL-D) operate in protected memory partitions with dedicated Hardware Memory Protection Units (MPU). Non-safety body functions operate in separate non-trusted partitions.

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Section 3: Software Components (SWCs) and Port Definitions
The Application Layer defines five primary software components:

3.1 ThrottleControl_SWC
- Component Type: Atomic Application Software Component (ASIL-D).
- Description: Responsible for calculating requested engine/motor throttle angle based on driver accelerator pedal inputs.
- Provided Ports:
  - PpThrottlePos: Provides calculated throttle command across interface If_ThrottlePosition.
- Required Ports:
  - RpPedalSensor: Receives dual redundant analog accelerator pedal sensor voltages.

3.2 BrakeManager_SWC
- Component Type: Atomic Application Software Component (ASIL-D).
- Description: Coordinates vehicle deceleration, brake-by-wire hydraulic blending, and regenerative torque requests.
- Provided Ports:
  - PpBrakeTorqueRequest: Provides electric motor regenerative torque request via interface If_BrakeTorque.
- Required Ports:
  - RpWheelSpeed: Receives four-wheel speed sensor pulses from the ABS/ESC controller.

3.3 BatteryManagement_SWC
- Component Type: Atomic Application Software Component (ASIL-C).
- Description: Monitors high-voltage battery state of charge (SoC), pack temperature, and high-voltage contactor state.
- Provided Ports:
  - PpBMSStatus: Broadcasts battery pack state and discharge limits via interface If_BatteryState.
- Required Ports:
  - RpCellVoltage: Receives individual lithium-ion cell voltages from internal BMS slave modules.

3.4 ThermalControl_SWC
- Component Type: Sensor/Actuator Software Component (ASIL-B).
- Description: Controls powertrain cooling loop water pumps, refrigerant valves, and front radiator fans.
- Provided Ports:
  - PpFanSpeedCommand: Actuates PWM radiator fan speed.
- Required Ports:
  - RpCoolantTemp: Reads coolant temperature sensors via interface If_ThermalData.
  - RpRefrigerantPressure: Reads high-pressure AC circuit pressure via interface If_RefrigerantPressure.

3.5 Gateway_SWC
- Component Type: Complex Device Driver / Application Gateway SWC (QM).
- Description: Translates and filters signals between CAN-FD Powertrain Bus and Automotive Ethernet Backbone.
- Provided Ports:
  - PpEthTelemetry: Transmits telemetry frames to vehicle telematics unit.

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Section 4: Software Interfaces and Signal Dictionary
All communications between SWCs use AUTOSAR Sender-Receiver interfaces.

4.1 Interface Definitions:
- Interface If_ThrottlePosition:
  - Sig_ThrottleAngle: float32, range [0.0, 100.0] %, Resolution 0.1 %, Update Rate: 10ms.
  - Sig_ThrottleFaultStatus: uint8, range [0, 3] (0=OK, 1=Degraded, 2=Fault, 3=SensorLoss), Update Rate: 10ms.
- Interface If_BrakeTorque:
  - Sig_RequestedTorque: int16, range [-500, 500] Nm, Resolution 1 Nm, Update Rate: 10ms.
  - Sig_BrakePedalState: boolean (0=Released, 1=Depressed), Update Rate: 10ms.
- Interface If_BatteryState:
  - Sig_StateOfCharge: uint8, range [0, 100] %, Resolution 1 %, Update Rate: 100ms.
  - Sig_PackVoltage: float32, range [0.0, 800.0] V, Resolution 0.05 V, Update Rate: 50ms.
  - Sig_PackCurrent: float32, range [-400.0, 400.0] A, Resolution 0.1 A, Update Rate: 20ms.
- Interface If_ThermalData:
  - Sig_CoolantTemp: int16, range [-40, 150] degC, Resolution 1 degC, Update Rate: 500ms.
  - Sig_PumpSpeedTarget: uint16, range [0, 6000] RPM, Resolution 10 RPM, Update Rate: 100ms.

4.2 Interface Table Summary:
Interface Name	Data Elements	Data Type	Cycle Time	Safety Level
If_ThrottlePosition	Sig_ThrottleAngle, Sig_ThrottleFaultStatus	float32, uint8	10ms	ASIL-D
If_BrakeTorque	Sig_RequestedTorque, Sig_BrakePedalState	int16, boolean	10ms	ASIL-D
If_BatteryState	Sig_StateOfCharge, Sig_PackVoltage, Sig_PackCurrent	uint8, float32, float32	100ms / 50ms	ASIL-C
If_ThermalData	Sig_CoolantTemp, Sig_PumpSpeedTarget	int16, uint16	500ms / 100ms	ASIL-B

Section 5: Functional Flows and Inter-SWC Communication
The system realizes vehicle operational behavior through coordinated functional execution chains:

5.1 Flow 1: Driver Acceleration Request Flow
1. Driver depresses the accelerator pedal.
2. Accelerator pedal voltage is acquired through MCAL ADC driver by RpPedalSensor port on ThrottleControl_SWC.
3. ThrottleControl_SWC validates dual sensor plausibility and checks for cross-channel plausibility faults.
4. Calculated throttle demand is published as Sig_ThrottleAngle on port PpThrottlePos.
5. The RTE transfers Sig_ThrottleAngle to the inverter actuator manager within 10ms.

5.2 Flow 2: Regenerative Braking Coordination Flow
1. BrakeManager_SWC detects brake pedal application and evaluates vehicle speed from RpWheelSpeed.
2. BrakeManager_SWC queries battery absorption capacity from BatteryManagement_SWC via PpBMSStatus.
3. BrakeManager_SWC calculates maximum permissible regenerative torque and issues torque request to InverterControl_SWC.
4. If battery State of Charge exceeds 95%, regenerative braking is tapered and hydraulic friction braking assumes full deceleration torque.

5.3 Flow 3: Thermal Protection Derating Flow
1. ThermalControl_SWC monitors Sig_CoolantTemp.
2. If coolant temperature exceeds 105 degC, ThermalControl_SWC requests power limit derating.
3. ThrottleControl_SWC clamps maximum allowable throttle angle to 50% to prevent electric motor stator overheating.

Section 6: Dependencies, Timing & RTE Interactions

This section specifies inter-module execution dependencies, periodic task mapping, and bus communications.

6.1 Task Scheduling Matrix

- Runnable_CalculateThrottle is mapped to Task_10ms.
- Runnable_EvaluateBrakeTorque is mapped to Task_10ms.
- Runnable_MonitorBatteryState is mapped to Task_20ms.
- Runnable_RegulateThermalLoop is mapped to Task_100ms.

6.2 Communication Timing Specifications

For network transmission on CAN-FD Channel 1, message frames are serialized by the AUTOSAR COM stack. The transmission period for Sig_ThrottleAngle is configured for 50ms cyclic execution on CAN-FD Channel 1 to conserve bus bandwidth.
(Note: Engineering review required: Compare with Section 4 update rate).

6.3 Fault Diagnostics and Recovery Concept

All hardware sensor anomalies and communication timeouts are logged via the Diagnostic Event Manager (DEM). For detailed diagnostic trouble code (DTC) mapping, see Section 9.4.
Software Component recovery actions follow ISO 26262 ASIL hazard mitigation guidelines.